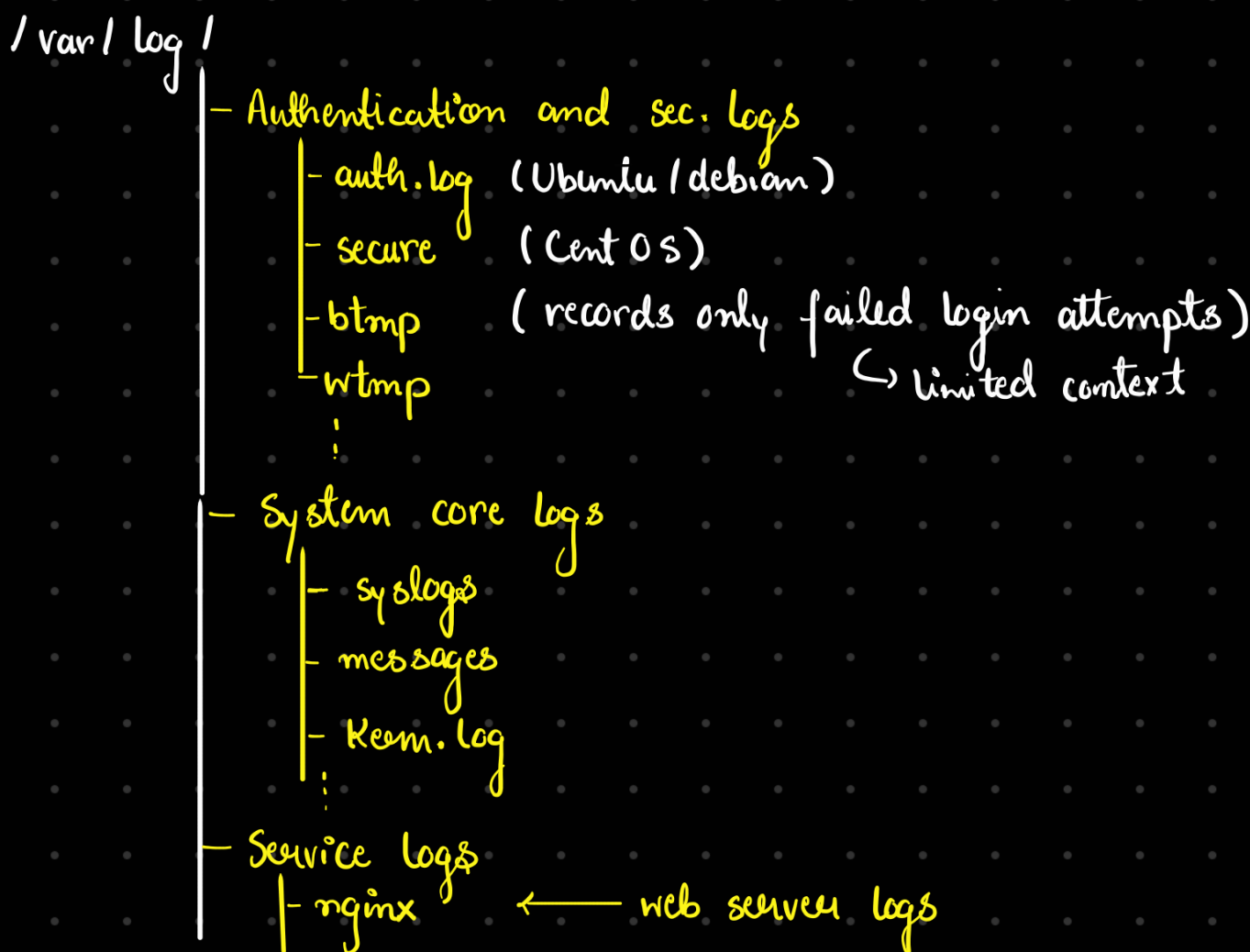
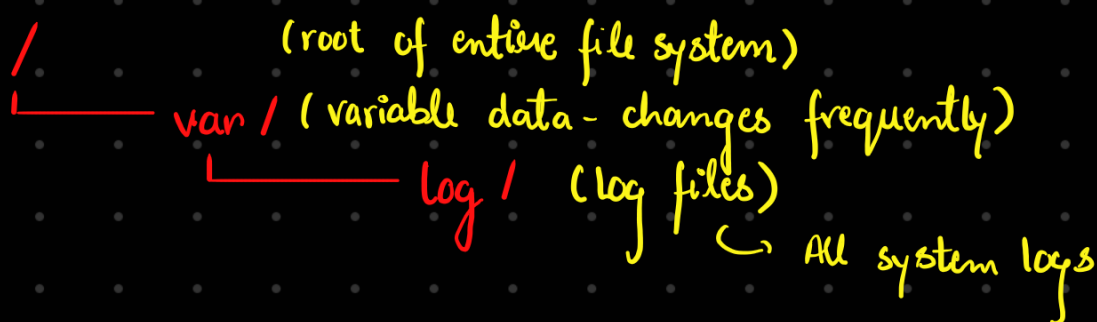


## Sentinel :

Python-based Linux security monitoring tool that analyzes SSH authentication logs, to detect unusual patterns and classify the authentication attacks based on their nature

## Architecture :

i) Log parsing : continuously tracks `/var/log/auth.log` directory within `/var/log` directory.



auth.log: Managed by - rsyslog daemon

Format - plain text, human readable

★ What gets logged?

: SSH auth. attempts, sudo command execution, pass. changes, acc. creation / deletion. ---

machine gen. the log

Timestamp Hostname Service [PID]: Message

e.g: Mar 13 10:30:15 server sshd[1234]: failed password for admin from 192.168.1.100

Mar ~~~~~ server sudo: --

" ~~~~~ server passwd[5678]: password changed for ubuntu.

★ Why not btmp? => doesn't record 'why login failed'  
↳ has binary format

Comparison: real time → btmp: binary, no real-time, less context  
→ auth.log: plain, real-time via, more context (tail -f)

→ helpful spirit

Daemon : A background process that runs cont. without direct user interaction. (started by system, runs cont. until system shutdown)

Foreground process : what we run / control

How daemons work ?

System boot :

- (i) Kernel loads
- (ii) init system starts (PID:1) // 1<sup>st</sup> process, parent of all
- (iii) init reads config. // init is old, modern version is systemd
- (iv) Daemons start in order
- (v) Daemons run continuously
- (vi) System shutdown

Systemd : (System & Service Manager) is a modern Daemon Manager version of old init system with faster & more features.

Systemd (1) — rsyslogd (456)  
— sshd (789)  
— bash (1234) — python (5678)

- ☆ Daemons are direct children of systemd.
- ☆ Our processes are .. .. over shell.

Daemon : implementation

Rsyslog : Rocket-fast system for log processing

purpose : centralized logging daemon  
log mssgs from programs

processes : filters, formats, routes programs

writes : To appropriate log files

## \* How connec<sup>n</sup> establishes from client - side :

- Ssh initialised
- 3 way TCP- handshake
- session key shared for safe data transfer (encryption)
- Auth. request is sent.

pluggable auth. module

## logging flow: program (sshd)

- receives auth. request ; is checked by PAM
- if there is unrecognized access then sshd takes action and  
↓  
generates log message: Failed password for "admin" from 192.168.1.100
- msg sent to rsyslog which reads configuration & carries the routing.

rsyslog daemon: receives message, reads config., determines destination based on facility & priority.

- rsyslog writes to /var/log/auth.log
- Sentinel reads auth.log

SSH in Windows : Exactly the same except the file locations.

Log parsing code -

```
from datetime import datetime
import re
import os

Failed_pass = re.compile(r'Failed password for(?:invalid user)
(?:P <user> \w+) from(?:P <ip> \S+)')
Conn"-closed = re.compile(r'Connection closed by(?:invalid user)
(?:P <user> \w+) \S+ (?:P <ip> \S+)')

def parse_auth_log(log_path = None):
    if not log_path:
        log_path = os.getenv("AUTH-LOG", "var/log/auth.log")
    events = [ ] # list of dictionaries
    with open(log_path, 'r') as file:
        for line in file: # iterable
            timestamp = None
            try:
                timestamp_str = line.split(None, 1)[0]
                if line.strip() else None
            if timestamp_str:
                timestamp = datetime.fromisoformat(
                    timestamp_str)
            except (ValueError, IndexError):
```

pass

m = Failed\_pass.search(line)

if m:

events.append({

'status': 'FAILED',

'user': m.group('user'),

'ip': m.group('ip')

'timestamp': timestamp

})

continue

n = Conn\_closed.search(line)

if n:

events.append({

'status': 'FAILED',

'user': m.group('user'),

'ip': m.group('ip')

'timestamp': timestamp

})

return events